Glossary

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National Center for Biotechnology Information

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Completed

Study to Describe the Safety, Tolerability, Immunogenicity, and Efficacy of RNA Vaccine Candidates Against COVID-19 in Healthy Individuals

ClinicalTrials.gov ID NCT04368728

Sponsor BioNTech SE

Information provided by BioNTech SE (Responsible Party)

Last Update Posted 2023-02-28

Study Details Tab

Study Overview

Brief Summary

This is a Phase 1/2/3, randomized, placebo-controlled, observer-blind, dose-finding, vaccine candidate-selection, and efficacy study in healthy individuals.

The study consists of 2 parts: Phase 1: to identify preferred vaccine candidate(s) and dose level(s); Phase 2/3: an expanded cohort and efficacy part.

The study will evaluate the safety, tolerability, and immunogenicity of 3 different SARS-CoV-2 RNA vaccine candidates against COVID-19 and the efficacy of 1 candidate:

- As a 2-dose (separated by 21 days) schedule;

- At various different dose levels in Phase 1;
- As a booster;
- In 3 age groups (Phase 1: 18 to 55 years of age, 65 to 85 years of age; Phase 2/3: ≥12 years of age [stratified as 12-15, 16-55 or >55 years of age]).

The candidate selected for efficacy evaluation in Phase 2/3 is BNT162b2 at a dose of 30 µg.

Participants who originally received placebo will be offered the opportunity to receive BNT162b2 at defined points as part of the study.

In order to describe the boostability of BNT162, and potential heterologous protection against emerging SARS-CoV-2 VOCs, an additional dose of BNT162b2 at 30 µg will be given to Phase 1 participants approximately 6 to 12 months after their second dose of BNT162b1 or BNT162b2. This will provide an early assessment of the safety of a third dose of BNT162, as well as its immunogenicity.

The assessment of boostability will be further expanded in a subset of Phase 3 participants at selected sites in the US who will receive a third dose of BNT162b2 at 30 µg or a third and potentially a fourth dose of prototype BNT162b2VOC at 30 µg (BNT162b2s01, based upon the South African variant and hereafter referred to as BNT162b2SA). A further subset of Phase 3 participants will receive a third, lower, dose of BNT162b2 at 5 or 10 µg.

To further describe potential homologous and heterologous protection against emerging SARS-CoV-2 VOCs, a new cohort of participants will be enrolled who are COVID-19 vaccine-naïve (ie, BNT162b2-naïve) and have not experienced COVID-19. They will receive BNT162b2SA given as a 2-dose series, separated by 21 days.

To reflect current and anticipated recommendations for COVID 19 vaccine boosters, participants in C4591001 who meet specified recommendations and have not already received one, will be offered a third dose of BNT162b2 after their second dose of BNT162.

Official Title

A PHASE 1/2/3, PLACEBO-CONTROLLED, RANDOMIZED, OBSERVER-BLIND, DOSE-FINDING STUDY TO EVALUATE THE SAFETY, TOLERABILITY, IMMUNOGENICITY, AND EFFICACY OF SARS-COV-2 RNA VACCINE CANDIDATES AGAINST COVID-19 IN HEALTHY INDIVIDUALS

Conditions

SARS-CoV-2 Infection COVID-19

Intervention / Treatment

- Biological: BNT162b1
- Biological: BNT162b2
- Other: Placebo
- Biological: BNT162b2SA

Other Study ID Numbers

- C4591001
- 2020-002641-42 (EudraCT Number)

Study Start (Actual)

2020-04-29

Primary Completion (Actual)

2023-02-10

Study Completion (Actual)

2023-02-10

Enrollment (Actual)

47079

Study Type

Interventional

Phase

Phase 2
Phase 3

Resource links provided by the National Library of Medicine

MedlinePlus (https://medlineplus.gov/) related topics: COVID-19 (Coronavirus Disease 2019) (https://medlineplus.gov/covid19coronavirusdisease2019.html) Post-COVID Conditions (Long COVID) (https://medlineplus.gov/postcovidconditionslongcovid.html)

FDA Drug and Device Resources (https://clinicaltrials.gov/fda-links)

Contacts and Locations

This section provides contact details for people who can answer questions about joining this study, and information on where this study is taking place.

To learn more, please see the Contacts and Locations section in How to Read a Study Record (https://clinicaltrials.gov/study-basics/how-to-read-study-record#contacts-and-locations).

This study has 169 locations

United States

Alabama Locations

- **Athens, Alabama, United States, 35611**
  North Alabama Research Center, LLC
- **Birmingham, Alabama, United States, 35216**
  Birmingham Clinical Research Unit
- **Huntsville, Alabama, United States, 35801**
  Medical Affiliated Research Center
- **Huntsville, Alabama, United States, 35802**
  Optimal Research, LLC
- **Mobile, Alabama, United States, 36608**
  Alliance for Multispecialty Research, LLC

Arizona Locations

- **Chinle, Arizona, United States, 86503**
  Chinle Comprehensive Health Care Facility
- **Chinle, Arizona, United States, 86503**
  Johns Hopkins Center for American Indian Health
- **Phoenix, Arizona, United States, 85018**
  The Pain Center of Arizona
- **Phoenix, Arizona, United States, 85023**
  HOPE Research Institute
- **Tempe, Arizona, United States, 85281**
  Alliance for Multispecialty Research, LLC
- **Whiteriver, Arizona, United States, 85941**
  Whiteriver Indian Hospital

California Locations

- **Anaheim, California, United States, 92801**
  Anaheim Clinical Trials, LLC
- **Long Beach, California, United States, 90806**
  Collaborative Neuroscience Research, LLC
- **Long Beach, California, United States, 90806**
  Long Beach Clinical Trials Services Inc.
- **Los Angeles, California, United States, 90027**
  Kaiser Permanente Los Angeles Medical Center
- **Los Angeles, California, United States, 90057**
  National Research Institute

North Hollywood, California, United States, 91606
Velocity Clinical Research, North Hollywood

Redding, California, United States, 96001
Paradigm Clinical Research Center

Sacramento, California, United States, 95815
Kaiser Permanente Sacramento

Sacramento, California, United States, 95817
Clinical and Translational Science Center (CTSC) Clinical Research Center (CCRC)

Sacramento, California, United States, 95817
UC Davis Medical Center

San Diego, California, United States, 92123-1881
California Research Foundation

Santa Clara, California, United States, 95051
Kaiser Permanente Santa Clara

Valley Village, California, United States, 91607
Bayview Research Group

Walnut Creek, California, United States, 94598
Diablo Clinical Research, Inc.

Colorado Locations

Aurora, Colorado, United States, 80012
Lynn Institute of Denver

Connecticut Locations

Milford, Connecticut, United States, 06460
Clinical Research Consulting, LLC

New Haven, Connecticut, United States, 06519
Yale Center for Clinical Investigations (CSRU)

Florida Locations

Coral Gables, Florida, United States, 33134
Alliance for Multispecialty Research

DeLand, Florida, United States, 32720
DeLand Clinical Research Unit

Fleming Island, Florida, United States, 32003
Fleming Island Center for Clinical Research

Hialeah, Florida, United States, 33012

Indago Research & Health Center, Inc

- **Hollywood, Florida, United States, 33024**
  Research Centers of America
- **Jacksonville, Florida, United States, 32216**
  Jacksonville Center for Clinical Research
- **Jacksonville, Florida, United States, 32256**
  Clinical Neuroscience Solutions, Inc.
- **Miami, Florida, United States, 33142**
  Acevedo Clinical Research Associates
- **Orlando, Florida, United States, 32801**
  Clinical Neuroscience Solutions, Inc. dba CNS Healthcare

Georgia Locations

- **Atlanta, Georgia, United States, 30331**
  Atlanta Center for Medical Research
- **Columbus, Georgia, United States, 31904**
  IACT Health
- **Savannah, Georgia, United States, 31406**
  Meridian Clinical Research, LLC
- **Stockbridge, Georgia, United States, 30281**
  Clinical Research Atlanta

Hawaii Locations

- **Honolulu, Hawaii, United States, 96814**
  East-West Medical Research Institute

Idaho Locations

- **Meridian, Idaho, United States, 83646**
  Solaris Clinical Research

Illinois Locations

- **Peoria, Illinois, United States, 61614**
  Optimal Research

Iowa Locations

- **Iowa City, Iowa, United States, 42242**
  University of Iowa Hospitals & Clinics
  Investigational Drug Servces
- **Iowa City, Iowa, United States, 52242**
  University of Iowa Hospitals & Clinics
- **Sioux City, Iowa, United States, 51106**
  Meridian Clinical Research, LLC

Kansas Locations

- **Newton, Kansas, United States, 67114**

Alliance for Multispecialty Research, LLC

Wichita, Kansas, United States, 67207
Alliance for Multispecialty Research, LLC

Kentucky Locations

Bardstown, Kentucky, United States, 40004
Kentucky Pediatric/ Adult Research

Louisiana Locations

Metairie, Louisiana, United States, 70006
Benchmark Research

New Orleans, Louisiana, United States, 70121
Ochsner Clinic Foundation

Shreveport, Louisiana, United States, 71101
LSU Health Sciences Center at Shreveport Clinical Trials Office

Shreveport, Louisiana, United States, 71103
LSUHSC-Shreveport

Maryland Locations

Baltimore, Maryland, United States, 21201
Pharmaron CPC, Inc.

Baltimore, Maryland, United States, 21201
University of Maryland Medical Center Investigational Drug Service Pharmacy

Baltimore, Maryland, United States, 21201
University of Maryland, Baltimore, Health Sciences Research Facility III

Baltimore, Maryland, United States, 21201
University of Maryland, Center for Vaccine Development and Global Health

Baltimore, Maryland, United States, 21224
Johns Hopkins Bayview Medical Center

Massachusetts Locations

Boston, Massachusetts, United States, 02118
Boston Medical Center

Worcester, Massachusetts, United States, 01655
UMass Memorial Medical Center - University Campus

Michigan Locations

Farmington Hills, Michigan, United States, 48334

Michigan Center for Medical Research

Mississippi Locations

Gulfport, Mississippi, United States, 39503
MedPharmics, LLC

Missouri Locations

Chesterfield, Missouri, United States, 63005
Clinical Research Professionals

St Louis, Missouri, United States, 63141
Sundance Clinical Research, LLC

Montana Locations

Bozeman, Montana, United States, 59715
Bozeman Health Deaconess Hospital dba Bozeman Health Clinical Research

Bozeman, Montana, United States, 59715
Bozeman Health Deaconess Hospital

Nebraska Locations

Fremont, Nebraska, United States, 68025
Methodist Physicians Clinic / CCT Research

Norfolk, Nebraska, United States, 68701
Meridian Clinical Research, LLC

Omaha, Nebraska, United States, 68114
Quality Clinical Research, Inc.

Omaha, Nebraska, United States, 68134
Meridian Clinical Research, LLC

Nevada Locations

Las Vegas, Nevada, United States, 89106
Wake Research-Clinical Research Center of Nevada, LLC

New Jersey Locations

Raritan, New Jersey, United States, 08869
Amici Clinical Research

Somers Point, New Jersey, United States, 08244
South Jersey Infectious Disease

New Mexico Locations

Gallup, New Mexico, United States, 87301
Johns Hopkins Center for American Indian Health

Shiprock, New Mexico, United States, 87420

Johns Hopkins Center for American Indian Health

New York Locations

Binghamton, New York, United States, 13901
Meridian Clinical Research, LLC

Endwell, New York, United States, 13760
Meridian Clinical Research LLC

New York, New York, United States, 10016
NYU Langone Health

New York, New York, United States, 10029
Icahn School of Medicine at Mount Sinai

Rochester, New York, United States, 14609
Rochester Clinical Research, Inc.

Rochester, New York, United States, 14642
University of Rochester Medical Center- Kari Steinmetz

Rochester, New York, United States, 14642
University of Rochester Medical Center

Syracuse, New York, United States, 13215
SUNY Upstate Medical University Global Health Research Unit

Vestal, New York, United States, 13850
Meridian Clinical Research LLC

North Carolina Locations

Cary, North Carolina, United States, 27518
PMG Research of Raleigh, LLC d/b/a PMG Research of Cary

Charlotte, North Carolina, United States, 28209
PMG Research of Charlotte LLC

Durham, North Carolina, United States, 27703
Accessioning Unit and Repository

Durham, North Carolina, United States, 27703
Duke Vaccine and Trials Unit

Durham, North Carolina, United States, 27705
Clinical Research Pickett Road

Durham, North Carolina, United States, 27710
Duke Investigational Drug Service Pharmacy

Greensboro, North Carolina, United States, 27408

PharmQuest

Hickory, North Carolina, United States, 28601
PMG Research of Hickory, LLC

Raleigh, North Carolina, United States, 27609
PMG Research of Raleigh, LLC

Raleigh, North Carolina, United States, 27612
M3 Wake Research, Inc.

Salisbury, North Carolina, United States, 28144
PMG Research of Salisbury, LLC

Wilmington, North Carolina, United States, 28401
PMG Research of Wilmington, LLC

Winston-Salem, North Carolina, United States, 27103
PMG Research of Winston-Salem, LLC

North Dakota Locations

Fargo, North Dakota, United States, 58104
Lillestol Research Llc

Ohio Locations

Cincinnati, Ohio, United States, 45219
Meridian Clinical Research, LLC

Cincinnati, Ohio, United States, 45229
Cincinnati Children's Hospital Medical Center

Cincinnati, Ohio, United States, 45246
Meridian Clinical Research LLC

Cleveland, Ohio, United States, 44106
University Hospitals Cleveland Medical Center

Cleveland, Ohio, United States, 44106
VA Northeast Ohio Healthcare System

Cleveland, Ohio, United States, 44122
Velocity Clinical Research, Inc.

Columbus, Ohio, United States, 43213
Aventiv Research Inc.

Dayton, Ohio, United States, 45406
Dayton Clinical Research

Dayton, Ohio, United States, 45409
Dayton Clinical Research

Dayton, Ohio, United States, 45419
PriMED Clinical Research

South Euclid, Ohio, United States, 44121

Senders Pediatrics

Oklahoma Locations

Norman, Oklahoma, United States, 73072
Lynn Institute of Norman

Oregon Locations

Portland, Oregon, United States, 97227
Kaiser Permanente Northwest-Center for Health Research

Pennsylvania Locations

Allentown, Pennsylvania, United States, 18102
Lehigh Valley Health Network/Network Office of Research and Innovation

Rhode Island Locations

East Greenwich, Rhode Island, United States, 02818
Velocity Clinical Research, Providence

South Carolina Locations

Little River, South Carolina, United States, 29566
Main Street Physician's Care

Loris, South Carolina, United States, 29569
Main Street Physician's Care

Tennessee Locations

Bristol, Tennessee, United States, 37620
Holston Medical Group

Kingsport, Tennessee, United States, 37660
Holston Medical Group

Knoxville, Tennessee, United States, 37909
Alliance for Multispecialty Research, LLC

Knoxville, Tennessee, United States, 37920
Alliance for Multispecialty Research, LLC

Memphis, Tennessee, United States, 38119
Clinical Neuroscience Solutions, Inc.

Nashville, Tennessee, United States, 37203
Clinical Research Associates, Inc.

Tullahoma, Tennessee, United States, 37388
Trinity Clinical Research

Texas Locations

Austin, Texas, United States, 78705

Benchmark Research

- **Austin, Texas, United States, 78726**
  ARC Clinical Research at Four Points
- **Austin, Texas, United States, 78745**
  Tekton Research, Inc.
- **Austin, Texas, United States, 78745**
  Tekton Research
- **Dallas, Texas, United States, 75246**
  North Texas Infectious Diseases Consultants, P.A.
- **Fort Worth, Texas, United States, 76104**
  Ventavia Research Group, LLC
- **Fort Worth, Texas, United States, 76135**
  Benchmark Research
- **Fort Worth, Texas, United States, 76135**
  Texas Health Resources
- **Galveston, Texas, United States, 77555**
  University of Texas Medical Branch
- **Houston, Texas, United States, 77008**
  Ventavia Research Group, LLC
- **Houston, Texas, United States, 77081**
  Texas Center for Drug Development, Inc.
- **Keller, Texas, United States, 76248**
  Ventavia Research Group, LLC
- **Mesquite, Texas, United States, 75149**
  SMS Clinical Research, LLC
- **Pearland, Texas, United States, 77584**
  LinQ Research, LLC
- **San Angelo, Texas, United States, 76904**
  Benchmark Research.
- **San Antonio, Texas, United States, 78229**
  Clinical Trials of Texas, Inc.
- **San Antonio, Texas, United States, 78229**
  Diagnostics Research Group
- **Tomball, Texas, United States, 77375**
  Martin Diagnostic Clinic

Utah Locations

- **Salt Lake City, Utah, United States, 84109**
  J. Lewis Research, Inc. / Foothill Family Clinic
- **Salt Lake City, Utah, United States, 84121**
  J. Lewis Research, Inc. / Foothill Family Clinic South

Virginia Locations

Annandale, Virginia, United States, 22003
Clinical Alliance for Research & Education - Infectious Diseases, LLC (CARE-ID)

Midlothian, Virginia, United States, 23114
Virginia Research Center LLC

Washington Locations

Seattle, Washington, United States, 98101
Benaroya Research Institute at Virginia Mason

Wenatchee, Washington, United States, 98801
Wenatchee Valley Hospital

Argentina

CABA, Argentina, 1426
Hospital Militar Central Cirujano Mayor Dr. Cosme Argerich

Brazil

São Paulo, Brazil, 04266-010
CEPIC - Centro Paulista de Investigacao Clinica e Servicos Medicos Ltda (Casa Branca)

Estado de Bahia Locations

Salvador, Estado de Bahia, Brazil, CEP: 40415-006
Hospital Santo Antonio/ Associacao Obras Sociais Irma Dulce

Germany

Berlin, Germany, 13353
CRS Clinical Research Services Berlin GmbH

Essen, Germany, 45355
Medizentrum Essen Borbeck

Frankfurt am Main, Germany, 60596
IKF Pneumologie GmbH & Co KG

Hamburg, Germany, 20359
Universitätsklinikum Hamburg-Eppendorf

Mannheim, Germany, 68167
CRS Clinical Research Services Mannheim GmbH

Stuhr, Germany, 28816
Studienzentrum Brinkum Dr. Lars Pohlmeier und Torsten Drescher

South Africa

Gauteng Locations

- **Johannesburg, Gauteng, South Africa, 2113**
  Newtown Clinical Research Centre
- **Pretoria, Gauteng, South Africa, 0183**
  Jongaie Research

Limpopo Locations

- **Thabazimbi, Limpopo, South Africa, 0380**
  Limpopo Clinical Research Initiative

Western Cape Locations

- **Cape Town, Western Cape, South Africa, 7530**
  Tiervlei Trial Centre, Basement Level, Karl Bremer Hospital

Turkey (Türkiye)

- **Ankara, Turkey (Türkiye), 06230**
  Ankara Universitesi Tip Fakultesi, Ibni Sina Hastanesi
- **Ankara, Turkey (Türkiye), 06230**
  Hacettepe Universitesi Tip Fakultesi
- **Istanbul, Turkey (Türkiye), 34020**
  Istanbul Yedikule Gogus Hastaliklari ve Gogus Cerrahisi Egitim Arastirma Hastanesi
- **Istanbul, Turkey (Türkiye), 34093**
  Istanbul Universitesi Istanbul Tip Fakultesi
- **Istanbul, Turkey (Türkiye), 34098**
  Istanbul Universitesi-Cerrahpasa, Cerrahpasa Tip Fakultesi
- **Istanbul, Turkey (Türkiye), 34214**
  Medipol Mega Universite Hastanesi
- **Istanbul, Turkey (Türkiye), 34303**
  Acibadem Atakent Hastanesi
- **Kocaeli, Turkey (Türkiye), 41380**
  Kocaeli Universitesi Tip Fakultesi
- **Sakarya, Turkey (Türkiye), 54100**
  Sakarya Universitesi Egitim ve Arastirma Hastanesi

Participation Criteria

Researchers look for people who fit a certain description, called eligibility criteria. Some examples of these criteria are a person's general health condition or prior treatments.

For general information about clinical research, read Learn About Studies (https://clinicaltrials.gov/study-basics/learn-about-studies).

Eligibility Criteria

Description

Inclusion Criteria:

• Male or female participants between the ages of 18 and 55 years, inclusive, 65 and 85 years, inclusive, or ≥12 years, inclusive, at randomization (dependent upon study phase). For the boostability and protection-against-VOCs subset: Existing participants enrolled to receive a third dose of BNT162b2 at 30 µg or BNT162b2SA; male or female participants between the ages of 18 and 55 years, inclusive, at rerandomization.

Newly enrolled participants enrolled to receive 2 doses of BNT162b2SA; male or female participants between the ages of 18 and 55 years, inclusive, at enrollment.

Existing participants enrolled to receive a third dose of BNT162b2 at 5 or 10 µg; male or female participants ≥18 years at rerandomization.

Note that participants <18 years of age cannot be enrolled in the EU.

- Participants who are willing and able to comply with all scheduled visits, vaccination plan, laboratory tests, lifestyle considerations, and other study procedures.
- Healthy participants who are determined by medical history, physical examination, and clinical judgment of the investigator to be eligible for inclusion in the study.
- Participants who, in the judgment of the investigator, are at risk for acquiring COVID-19.
- Boostability and protection-against-VOCs existing participant subset only: Participants who provided a serum sample at Visit 3, with Visit 3 occurring within the protocol-specified window.
- Capable of giving personal signed informed consent

Exclusion Criteria:

- Other medical or psychiatric condition including recent (within the past year) or active suicidal ideation/behavior or laboratory abnormality that may increase the risk of study participation or, in the investigator's judgment, make the participant inappropriate for the study.
- Phases 1 and 2 only: Known infection with human immunodeficiency virus (HIV), hepatitis C virus (HCV), or hepatitis B virus (HBV).
- History of severe adverse reaction associated with a vaccine and/or severe allergic reaction (eg, anaphylaxis) to any component of the study intervention(s).
- Receipt of medications intended to prevent COVID 19.
- Previous clinical (based on COVID-19 symptoms/signs alone, if a SARS-CoV-2 NAAT result was not available) or microbiological (based on COVID-19 symptoms/signs and a positive SARS-CoV-2 NAAT result) diagnosis of COVID 19

- Phase 1 only: Individuals at high risk for severe COVID-19, including those with any of the following risk factors:

    - Hypertension

  - Diabetes mellitus
  - Chronic pulmonary disease
  - Asthma
  - Current vaping or smoking
  - History of chronic smoking within the prior year
  - BMI >30 kg/m2
  - Anticipating the need for immunosuppressive treatment within the next 6 months
- Phase 1 only: Individuals currently working in occupations with high risk of exposure to SARS-CoV-2 (eg, healthcare worker, emergency response personnel).
- Immunocompromised individuals with known or suspected immunodeficiency, as determined by history and/or laboratory/physical examination.
- Phase 1 only: Individuals with a history of autoimmune disease or an active autoimmune disease requiring therapeutic intervention.
- Bleeding diathesis or condition associated with prolonged bleeding that would, in the opinion of the investigator, contraindicate intramuscular injection.
- Women who are pregnant or breastfeeding.
- Previous vaccination with any coronavirus vaccine.
- Individuals who receive treatment with immunosuppressive therapy, including cytotoxic agents or systemic corticosteroids, eg, for cancer or an autoimmune disease, or planned receipt throughout the study.
- Phase 1 only: Regular receipt of inhaled/nebulized corticosteroids.
- Receipt of blood/plasma products or immunoglobulin, from 60 days before study intervention administration or planned receipt throughout the study.
- Participation in other studies involving study intervention within 28 days prior to study entry through and including 6 months after the last dose of study intervention, with the exception of non-Pfizer interventional studies for prevention of COVID 19, which are prohibited throughout study participation.
- Previous participation in other studies involving study intervention containing lipid nanoparticles.
- Phase 1 only: Positive serological test for SARS-CoV-2 IgM and/or IgG antibodies at the screening visit.
- Phase 1 only: Any screening hematology and/or blood chemistry laboratory value that meets the definition of a ≥ Grade 1 abnormality.
- Phase 1 only: Positive test for HIV, hepatitis B surface antigen (HBsAg), hepatitis B core antibodies (HBc Abs), or hepatitis C virus antibodies (HCV Abs) at the screening visit.
- Phase 1 only: SARS-CoV-2 NAAT-positive nasal swab within 24 hours before receipt of study intervention.
- Investigator site staff or Pfizer employees directly involved in the conduct of the study, site staff otherwise supervised by the investigator, and their respective family members.

Ages Eligible for Study

12 Years and older (Child, Adult, Older Adult)

Sexes Eligible for Study

All

Accepts Healthy Volunteers

Yes

Study Plan

This section provides details of the study plan, including how the study is designed and what the study is measuring.

How is the study designed?

Design Details

Primary Purpose : Prevention
Allocation : Randomized
Interventional Model : Parallel Assignment
Masking : Triple (Participant, Care Provider, Investigator)

Arms and Interventions

Participant Group/Arm	Intervention/Treatment
Experimental: 10 µg dose, 18-55 years of age (2 doses)	Biological: BNT162b1 • Intramuscular injection Biological: BNT162b2 • Intramuscular injection
Experimental: 20 µg dose, 18-55 years of age (2 doses)	Biological: BNT162b1 • Intramuscular injection Biological: BNT162b2 • Intramuscular injection
Experimental: 30 µg dose, 18-55 years of age (2 doses)	Biological: BNT162b1 • Intramuscular injection Biological: BNT162b2 • Intramuscular injection
Experimental: 10 µg dose, 65-85 years of age (2 doses)	Biological: BNT162b1 • Intramuscular injection Biological: BNT162b2 • Intramuscular injection
Experimental: 20 µg dose, 65-85 years of age (2 doses)	Biological: BNT162b1 • Intramuscular injection Biological: BNT162b2

	• Intramuscular injection
Experimental: 30 µg dose, 65-85 years of age (2 doses)	Biological: BNT162b1 • Intramuscular injection Biological: BNT162b2 • Intramuscular injection
Experimental: 30 µg dose, ≥12 years of age (2 doses)	Biological: BNT162b2 • Intramuscular injection
Placebo Comparator: Placebo, 18-55 years of age	Other: Placebo • Intramuscular injection
Placebo Comparator: Placebo, 65-85 years of age	Other: Placebo • Intramuscular injection
Placebo Comparator: Placebo, ≥12 years of age	Other: Placebo • Intramuscular injection
Experimental: 100 µg dose, 18-55 years of age (2 doses)	Biological: BNT162b1 • Intramuscular injection
Other: Vaccination of Placebo recipients with BNT162b2 - Stage 1 Participants ≥16 years of age who originally	Biological: BNT162b2 • Intramuscular injection

received placebo and are eligible for COVID-19 vaccination following any local or national recommendations will be offered the opportunity to receive BNT162b2 as part of the study.	
Other: Vaccination of placebo recipients with BNT162b2 - Stage 2 Participants ≥16 years of age who originally received placebo will be offered the opportunity to receive BNT162b2 at defined points as part of the study.	Biological: BNT162b2 • Intramuscular injection
Experimental: Booster vaccination of Phase 1 participants with BNT162b2 at a dose of 30 µg	Biological: BNT162b2 • Intramuscular injection
Experimental: Booster vaccination of Phase 3 participants with BNT162b2 at a dose of 30 µg	Biological: BNT162b2 • Intramuscular injection
Experimental: Booster vaccination of Phase 3 participants with	Biological: BNT162b2SA • Intramuscular injection

BNT162b2SA at a dose of 30 µg	
Experimental: Vaccination of BNT162b2-naive participants with BNT162b2SA at a dose of 30 µg	Biological: BNT162b2SA • Intramuscular injection
Experimental: Booster and further vaccination of Phase 3 participants with BNT162b2SA at a dose of 30 µg	Biological: BNT162b2SA • Intramuscular injection
Experimental: Booster vaccination of Phase 3 participants with BNT162b2 at a dose of 5 µg	Biological: BNT162b2 • Intramuscular injection
Experimental: Booster vaccination of Phase 3 participants with BNT162b2 at a dose of 10 µg	Biological: BNT162b2 • Intramuscular injection

What is the study measuring?

Primary Outcome Measures

Outcome Measure	Measure Description	Time Frame
Percentage of participants in Phase 1	Pain at the injection site, redness, and swelling as self-reported on electronic diaries.	For 7 days after dose 1

reporting local reactions		and dose 2
Percentage of participants in Phase 1 reporting systemic events	Fever, fatigue, headache, chills, vomiting, diarrhea, new or worsened muscle pain, and new or worsened joint pain as self-reported on electronic diaries.	For 7 days after dose 1 and dose 2
Percentage of participants in Phase 1 reporting adverse events	As elicited by investigational site staff	From dose 1 through 1 month after the last dose
Percentage of participants in Phase 1 reporting serious adverse events	As elicited by investigational site staff	From dose 1 through 6 months after the last dose
Percentage of Phase 1 participants with abnormal hematology and chemistry laboratory values	As measured at the central laboratory	1 day after dose 1

Percentage of Phase 1 participants with abnormal hematology and chemistry laboratory values	As measured at the central laboratory	7 days after dose 1
Percentage of Phase 1 participants with abnormal hematology and chemistry laboratory values	As measured at the central laboratory	7 days after dose 2
Percentage of Phase 1 participants with grading shifts in hematology and chemistry laboratory assessments	As measured at the central laboratory	Between baseline and 1 day after dose 1
Percentage of Phase 1 participants with grading shifts in hematology and chemistry laboratory assessments	As measured at the central laboratory	Between baseline and 7 days after dose 1

Percentage of Phase 1 participants with grading shifts in hematology and chemistry laboratory assessments	As measured at the central laboratory	Between before dose 2 and 7 days after dose 2
In the first 360 participants randomized into Phase 2/3, percentage of participants reporting local reactions	Pain at the injection site, redness, and swelling as self-reported on electronic diaries.	For 7 days after dose 1 and dose 2
In the first 360 participants randomized into Phase 2/3, percentage of participants reporting systemic events	Fever, fatigue, headache, chills, vomiting, diarrhea, new or worsened muscle pain, and new or worsened joint pain as self-reported on electronic diaries.	For 7 days after dose 1 and dose 2
In the first 360 participants randomized into Phase 2/3, percentage of participants reporting adverse events	As elicited by investigational site staff	From dose 1 through 1 month after the last dose

In the first 360 participants randomized into Phase 2/3, percentage of participants reporting serious adverse events	As elicited by investigational site staff	From dose 1 through 6 months after the last dose
In a subset of at least 6000 participants randomized in Phase 2/3, percentage of participants reporting local reactions	Pain at the injection site, redness, and swelling as self-reported on electronic diaries.	For 7 days after dose 1 and dose 2
In a subset of at least 6000 participants randomized in Phase 2/3, percentage of participants reporting systemic events	Fever, fatigue, headache, chills, vomiting, diarrhea, new or worsened muscle pain, and new or worsened joint pain as self-reported on electronic diaries.	For 7 days after dose 1 and dose 2
Percentage of participants in Phase 2/3 reporting adverse events	As elicited by investigational site staff	From dose 1 through 1 month after the last dose

Percentage of participants in Phase 2/3 reporting serious adverse events	As elicited by investigational site staff	From dose 1 through 6 months after the last dose
Confirmed COVID-19 in Phase 2/3 participants without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 7 days after the second dose of study interve ntion to the end of the study, up to 2 years
Confirmed COVID-19 in Phase 2/3 participants with and without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 7 days after the second dose of study interve ntion to the end of the study, up to 2 years

Percentage of participants 12-15 years of age in Phase 3 reporting adverse events	As elicited by investigational site staff	From dose 1 through 1 month after the last dose
Percentage of participants 12-15 years of age in Phase 3 reporting adverse events	As elicited by investigational site staff	From dose 1 through 6 months after the last dose
In participants 12-15 years of age randomized in Phase 3, percentage of participants reporting local reactions	Pain at the injection site, redness, and swelling as self-reported on electronic diaries.	For 7 days after dose 1 and dose 2
In participants 12-15 years of age randomized in Phase 3, percentage of participants reporting systemic events	Fever, fatigue, headache, chills, vomiting, diarrhea, new or worsened muscle pain, and new or worsened joint pain as self-reported on electronic diaries.	For 7 days after dose 1 and dose 2

In participants who receive BNT162b2SA given as 1 or 2 doses, percentage of participants reporting adverse events	As elicited by investigational site staff	From dose 1 through 1 month after the last dose
In participants who receive BNT162b2SA given as 1 or 2 doses, percentage of participants reporting serious adverse events	As elicited by investigational site staff	From dose 1 through 5 or 6 months after the last dose
In participants, who receive BNT162b2SA given as 1 or 2 doses, percentage of participants reporting local reactions	Pain at the injection site, redness, and swelling as self-reported on electronic diaries.	For 7 days after dose 1 (and dose 2)
In participants who receive BNT162b2SA given as 1 or 2 doses, percentage of participants reporting	Fever, fatigue, headache, chills, vomiting, diarrhea, new or worsened muscle pain, and new or worsened joint pain as self-reported on electronic diaries.	For 7 days after dose 1 (and dose 2)

systemic events		
In participants who receive a third dose of BNT162b2 as part of the subset for evaluation of boostability and protection against emerging VOCs, percentage of participants reporting adverse events	As elicited by investigational site staff	From the third dose through 1 month after the third dose
In participants who receive a third dose of BNT162b2 as part of the subset for evaluation of boostability and protection against emerging VOCs, percentage of participants reporting serious adverse events	As elicited by investigational site staff	From the third dose through 6 months after the third dose
In participants who receive a	Pain at the injection site, redness, and swelling as self-reported on electronic	For 7 days

third dose of BNT162b2 as part of the subset for evaluation of boostability and protection against emerging VOCs, percentage of participants reporting local reactions	diaries.	after the third dose
In participants who receive a third dose of BNT162b2 as part of the subset for evaluation of boostability and protection against emerging VOCs, percentage of participants reporting systemic events	Fever, fatigue, headache, chills, vomiting, diarrhea, new or worsened muscle pain, and new or worsened joint pain as self-reported on electronic diaries.	For 7 days after the third dose
In participants who receive a third dose of BNT162b2 as a result of current or anticipated	As elicited by investigational site staff	From the third dose through 1 month

recommendations, percentage of participants reporting adverse events		after the third dose
In participants who receive a third dose of BNT162b2 as a result of current or anticipated recommendations, percentage of participants reporting serious adverse events	As elicited by investigational site staff	From the third dose through 6 months after the third dose
Noninferiority of the SARS-CoV-2 reference strain neutralizing titers after a third dose of BNT162b2 at 30 µg compared to after 2 doses of BNT162b2, in the same individuals	As measured at the central laboratory	1 month after the third dose
Noninferiority of the SARS-	As measured at the central laboratory	1 month

CoV-2 SA strain neutralizing titers after one dose of BNT162b2SA compared to the SARS-CoV-2 reference strain neutralizing titers after 2 doses of BNT162b2, in the same individuals		after the third dose
Noninferiority of the SARS-CoV-2 SA strain neutralizing titers after 2 doses of BNT162b2SA compared to the SARS-CoV-2 reference strain neutralizing titers after 2 doses of BNT162b2	As measured at the central laboratory	1 month after the second dose

Secondary Outcome Measures

Outcome Measure	Measure Description	Time Frame
In Phase 1 participants,	As measured at the central laboratory	Throug h 2

SARS-CoV-2 serum neutralizing antibody levels, expressed as GMTs		years after the final dose
In Phase 1 participants, GMFR in SARS-CoV-2 serum neutralizing titers from before vaccination to each subsequent time point	As measured at the central laboratory	Through 2 years after the final dose
Proportion of participants in Phase 1 achieving a greater than or equal to 4-fold rise from before vaccination in SARS-CoV-2 serum neutralizing antibody levels	As measured at the central laboratory	Through 2 years after the final dose
In Phase 1 participants, SARS-CoV-2 anti-S1 binding antibody levels	As measured at the central laboratory	Through 2 years after the

and anti-RBD binding antibody levels, expressed as GMCs		final dose
Proportion of participants in Phase 1 achieving a greater than or equal to 4-fold rise from before vaccination in SARS-CoV-2 anti-S1 binding antibody levels and anti-RBD binding antibody levels	As measured at the central laboratory	Through 2 years after the final dose
In Phase 1 participants, GMFR in SARS-CoV-2 anti-S1 binding antibody levels and anti-RBD binding antibody levels from before vaccination to each subsequent time point	As measured at the central laboratory	Through 2 years after the final dose
In Phase 1 participants,	As measured at the central laboratory	Through 2

GMR of the geometric mean of SARS-CoV-2 serum neutralizing titers to the geometric mean of SARS CoV 2 (anti-S1 and anti-RBD) binding antibody levels		years after the final dose
Confirmed COVID-19 in Phase 2/3 participants without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 14 days after the second dose of study interve ntion to the end of the study, up to 2 years
Confirmed COVID-19 in Phase 2/3 participants with and without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 14 days after the second dose of study interve ntion to the end

		of the study, up to 2 years
Confirmed severe COVID-19 in Phase 2/3 participants without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 7 days after the second dose of study interve ntion to the end of the study, up to 2 years
Confirmed severe COVID-19 in Phase 2/3 participants without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 14 days after the second dose of study interve ntion to the end of the study, up to 2 years
Confirmed severe COVID-19 in Phase 2/3	Per 1000 person-years of follow-up	From 7 days after the

participants with and without evidence of infection before vaccination		second dose of study interve ntion to the end of the study, up to 2 years
Confirmed severe COVID-19 in Phase 2/3 participants with and without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 14 days after the second dose of study interve ntion to the end of the study, up to 2 years
Confirmed COVID-19 (according to the CDC-defined symptoms) in Phase 2/3 participants without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 7 days after the second dose of study interve ntion to the end of the study,

		up to 2 years
Confirmed COVID-19 (according to the CDC-defined symptoms) in Phase 2/3 participants without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 14 days after the second dose of study intervention to the end of the study, up to 2 years
Confirmed COVID-19 (according to the CDC-defined symptoms) in Phase 2/3 participants with and without evidence of infection before vaccination	Per 1000 person-years of follow-up	From 7 days after the second dose of study intervention to the end of the study, up to 2 years
Confirmed COVID-19 (according to the CDC-defined symptoms) in	Per 1000 person-years of follow-up	From 14 days after the second

Phase 2/3 participants with and without evidence of infection before vaccination		dose of study interve ntion to the end of the study, up to 2 years
GMR of SARS CoV 2 neutralizing titers in the 2 age groups (12-15 years of age to 16-25 years of age)	As measured at the central laboratory	1 month after the second dose
Incidence of asymptomatic SARS CoV-2 infection based on N binding antibody seroconversio n in participants with no serological or virological evidence of past SARS CoV-2 infection or confirmed COVID-19 prior to 1 month after receipt of	Per 1000 person-years of follow-up	Throug h 1 month after the second dose

the second dose		
Incidence of asymptomatic SARS CoV-2 infection based on central laboratory-confirmed NAAT in participants with no serological or virological evidence (up to the start of the asymptomatic surveillance period) of past SARS-CoV-2 infection	Per 1000 person-years of follow-up	Through 6 months after the second dose
Noninferiority of the SARS-CoV-2 SA strain neutralizing titers after a third dose of BNT162b2 at 30 µg compared to the SARS-CoV-2 reference strain neutralizing titers after 2	As measured at the central laboratory	1 month after the third dose

doses of BNT162b2, in the same individuals		
Noninferiority of the SARS-CoV-2 reference strain neutralizing titers after one dose of BNT162b2SA compared to after 2 doses of BNT162b2, in the same individuals	As measured at the central laboratory	1 month after the first dose of BNT162b2SA
Comparison of the SARS-CoV-2 SA strain neutralizing titers after 1 dose of BNT162b2SA to after a third dose of BNT162b2 at 30 µg	As measured at the central laboratory	1 month after the first dose of BNT162b2SA/ third dose of BNT162b2

Comparison of the SARS-CoV-2 SA strain neutralizing titers after 2 doses of BNT162b2SA to the SARS-CoV-2 reference strain neutralizing titers after 2 doses of BNT162b2, in the same individuals	As measured at the central laboratory	1 month after the second dose of BNT162b2SA
Comparison of the SARS-CoV-2 SA strain neutralizing titers after 2 doses of BNT162b2SA to after 2 doses of BNT162b2	As measured at the central laboratory	1 month after the second dose

Comparison of the SARS-CoV-2 reference strain neutralizing titers after 2 doses of BNT162b2SA to after 2 doses of BNT162b2	As measured at the central laboratory	1 month after the second dose

Collaborators and Investigators

This is where you will find people and organizations involved with this study.

Sponsor

BioNTech SE

Collaborators

- Pfizer

Investigators

- Study Director: Pfizer CT.gov Call Center, Pfizer

Publications

From PubMed

These publications are automatically filled in from PubMed, a public database of scientific and medical articles, and may or may not be about the study.

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Study Record Dates

These dates track the progress of study record and summary results submissions to ClinicalTrials.gov. Study records and reported results are reviewed by the National Library of Medicine (NLM) to make sure they meet specific quality control standards before being posted on the public website.

Study Registration Dates

First Submitted

2020-04-27

First Submitted that Met QC Criteria

2020-04-29

First Posted

2020-04-30

Study Record Updates

Last Update Submitted that Met QC Criteria

2023-02-27

Last Update Posted

2023-02-28

Last Verified

2023-02

More Information

Terms related to this study

Keywords Provided by BioNTech SE

COVID-19
Coronavirus
Vaccine
SARS-CoV-2
RNA Vaccine

Additional Relevant MeSH Terms

Pneumonia, Viral

Pneumonia
Respiratory Tract Infections
Infections
Virus Diseases
Coronaviridae Infections
Nidovirales Infections
RNA Virus Infections
Lung Diseases
Respiratory Tract Diseases
COVID-19
Coronavirus Infections
mRNA Vaccines
Nucleic Acid-Based Vaccines
Vaccines, Synthetic
Recombinant Proteins
Proteins
Amino Acids, Peptides, and Proteins
Vaccines
Biological Products
Complex Mixtures
COVID-19 Vaccines
Viral Vaccines
Antigens
Biological Factors
BNT162 Vaccine

Plan for Individual Participant Data (IPD)

Plan to Share Individual Participant Data (IPD)?

No

Drug and device information, study documents, and helpful links

Studies a U.S. FDA-Regulated Drug Product

Yes

Studies a U.S. FDA-Regulated Device Product

No

Helpful Links Provided by BioNTech SE

To obtain contact information for a study center near you, click here.
(https://pmiform.com/clinical-trial-info-request?StudyID=C4591001)